

BISFIT

Retrieval-Augmented Generation for Bureau of Indian Standards

JUSTIFICATION DOCUMENT

Why the Problem Was Chosen · The Gap It Addresses · Expected

Why This Problem Was Chosen

India's construction sector contributes ~9% of GDP and is governed by 580 Indian Standards covering every building material — cement, steel, timber, glass, sanitary fittings, and more. Compliance with these IS codes is a statutory requirement under the BIS Act, 2016 and the National Building Code of India. Yet the primary reference for these standards, SP 21 : 2005, exists only as a dense multi-column PDF that standard tools cannot reliably parse.

A site engineer trying to verify compressive strength requirements for 53 Grade OPC, or a QA inspector confirming dimensional tolerances for hollow concrete blocks, must manually search hundreds of pages — a process that takes 45–90 minutes per query and depends entirely on prior IS code knowledge. For small contractors, manufacturers seeking ISI certification, and field inspectors in Tier 2 and Tier 3 cities, this barrier is effectively insurmountable.

The problem was chosen because it is simultaneously high-stakes — non-compliance contributes to structural failures and legal liability — and technically tractable: the information already exists in SP 21 : 2005. The gap is one of retrieval and accessibility, not information scarcity.

The Gap BISfit Addresses

Existing tools fail for four compounding reasons that BISfit resolves directly:

- PDF extraction failure: multi-column BIS layouts produce interleaved, semantically broken text with standard parsers. BISfit's parser.py uses PyMuPDF with section-aware block detection to extract clean, per-standard text.
- Chunking mismatch: sliding-window chunking fragments individual IS standards mid-clause. BISfit uses domain-aware chunking — one IS standard = one semantic chunk — preserving every standard as a self-contained retrievable unit.
- Vocabulary mismatch: practitioners query in plain English; IS documents use technical clause language. BISfit resolves this through metadata-prefixed embeddings (IS number inside the vector) and Llama 3.1 8b query reformulation, so both 'IS 383' and 'coarse aggregate for structural concrete' hit the same chunk.
- Hallucination risk: LLMs without grounding fabricate IS codes. BISfit generates answers exclusively from retrieved SP 21 chunks, with a regex fallback ensuring no response is ever untraced to a source document.

Expected Real-World Impact

BISfit collapses a 45–90 minute compliance search into a sub-2-second query — validated by measured performance across a public benchmark:

Metric	Result
Hit Rate @ 3 (target > 80%)	✓ 90%
MRR @ 5 (target > 0.70)	✓ 0.90
Avg. Latency (target < 5.0 s)	✓ 1.71 s

For site engineers and QA inspectors, this means real-time on-site compliance checks before material acceptance — a structural safety checkpoint that currently does not exist at scale. For manufacturers, it removes a key barrier to ISI certification. For architects and contractors, it enables cited, traceable specifications at the point of design.

Beyond individual productivity, BISfit democratises access to standards knowledge that today is practically available only to large organisations with dedicated technical teams — extending it to every professional who builds India, regardless of city, firm size, or IS code fluency.